

PedLabel

Pedestrian Image Annotation Platform

UX Project Brief

Pascal Musabyimana · April 2026 · UX Design Lab

1 · Define

Problem Statement

Annotating pedestrian datasets is a mess. Most teams use 2+ disconnected tools — one for bounding boxes, another for keypoints, something else for exports — and none of them talk to each other. The result: inconsistent labels, export format headaches, and way too much time debugging tooling instead of doing actual research.

No lightweight, research-focused tool currently combines bounding boxes, keypoint marking, occlusion tagging, and behavioral attribute labeling in a single interface — alongside the quality control features an academic lab actually needs: inter-annotator agreement tracking, review queues, and label versioning.

User Personas

Persona	Role	Goal	Frustration
Alex Chen, 26 — PhD Researcher (Primary)	PhD student, Computer Vision lab	Build & annotate a 15,000-image pedestrian dataset for thesis research on crowd behaviour. High tech level: Python, PyTorch, CLI.	Export formats never match training pipelines. Spends more time on tooling than research.
Maria Lopes, 33 — Lab Coordinator	Research lab coordinator / annotation team lead	Manage 6 annotators, enforce label quality, track progress toward dataset milestones.	No visibility into annotator struggles. Tracks progress in a separate spreadsheet.
Tom Baert, 21 — Undergraduate Annotator	Undergraduate research assistant, part-time annotator	Label assigned image batches accurately and efficiently with no ML background.	Unclear guidelines, complex UIs. Unsure whether labels are good enough.

User Journey Map

Stage	Actor	Action	Pain Point
1. Project Setup	Alex	Creates project, defines label schema, uploads images	Rigid schema editors — adding attributes mid-project breaks existing labels
2. Assignment	Maria	Assigns batches to annotators, sets deadline	No built-in assignment tool — done via email and spreadsheet

Stage	Actor	Action	Pain Point
3. Annotation	Tom	Draws bounding boxes, tags attributes, submits batch	Too many tool modes; keyboard shortcut discovery is poor
4. QA Review	Maria	Reviews flagged labels, accepts/rejects, sends feedback	No inline feedback — reviewers must email comments separately
5. Export	Alex	Exports dataset in COCO JSON for model training	Format conversion errors; field naming differences cause debugging hours

Value Proposition

PedLabel is a free, research-focused annotation platform for academic teams — bounding boxes, pose keypoints, occlusion flags, and behavioral tagging in one interface, with built-in QA workflows and one-click export to COCO JSON, YOLO TXT, and Pascal VOC XML. No DevOps setup, no enterprise price tag. Unlike paid enterprise platforms, PedLabel is designed specifically for small-to-medium academic teams.

Project Scope

In scope for v1:

- Image & video frame annotation
- Bounding box + pose keypoint tools (17-point COCO skeleton)
- Occlusion and attribute tagging (age group, activity, visibility)
- Annotator assignment & workload management
- QA review queue — accept / reject / comment workflow
- Inter-annotator agreement statistics
- Export to COCO JSON, YOLO TXT, Pascal VOC XML
- Role-based auth (Researcher / Coordinator / Annotator)

Not included (v1):

- Mobile app, real-time collaborative editing, AI-assisted auto-labeling
- Model training / inference, enterprise billing

2 · Research

Competitor Analysis

Tool	Pedestrian Features	Collab	Cost	Key Weakness
CVAT	BBox, polygon, keypoints	Yes	Free	Steep learning curve, complex setup
Labellmg	BBox only	No	Free	No video, no QA, no collaboration
Roboflow	BBox, polygon	Limited	Freemium	No keypoints on free tier, image limits

User Testing Takeaways

Informal sessions with 3 lab members on CVAT and Roboflow:

- Active class got lost in the sidebar — needs to be permanently visible.
- QA review was unclear reviewers had to navigate elsewhere to see why a label was flagged.
- Keyboard shortcuts sped things up massively but nobody knew they existed — all three asked for a persistent shortcut overlay.
- Users wanted a visual progress bar per batch, not just a number.
- Export failed silently when field names didn't match COCO schema — error messages were useless.
- Student annotator repeatedly used the wrong class because label descriptions weren't accessible inline.

Survey Results

Short survey, 7 questions, n=23 (computer vision course cohort):

Finding	Result
Use 2+ tools in a single workflow	74%
Biggest frustration: inconsistent team annotations	61%
Biggest frustration: bad export compatibility	43%
Have no formal QA step	65%
Rate keyboard shortcuts as important / very important	87%
Would switch to a free research-focused tool	78%
Most-missing: inter-annotator agreement stats	48%
Most-missing: inline QA comments	39%

3 · Analyse

User Stories

- As a researcher, I want to create a project with a custom label schema so my team annotates consistently from day one.
- As an annotator, I want bounding boxes and attribute tagging in one interface so I don't switch tools mid-session.
- As a coordinator, I want to see each annotator's progress and review flagged labels in a QA queue before issues pile up.
- As a researcher, I want to export in COCO JSON with a validation report so I can plug it straight into my training pipeline.
- As an annotator, I want visible keyboard shortcuts for all common actions so I can annotate faster without reaching for the mouse.
- As a coordinator, I want inter-annotator agreement stats per class so I know which labels need clearer guidelines.

User Flows

Flow 1 — Project Setup (Researcher)

- Login → 'New Project' → 3-step wizard opens (modal overlay on blurred dashboard)
- Enter name, description, dataset size — live draft summary updates as fields are filled
- Define label schema: classes, colours, descriptions, example images
- Upload images (ZIP or folder) → progress bar
- Assign batches to annotators, set deadline + quality threshold
- Project goes live → dashboard shows 0% progress

Flow 2 — Annotation Session (Annotator)

- Login → see assigned batches with progress bars
- Open batch → image viewer loads; active class shown in top bar
- Draw bounding box (click-drag) → attribute panel slides in
- Tag: age group, occlusion, activity → press N to confirm & advance
- Complete batch → Submit → enters QA queue

Flow 3 — QA Review (Coordinator)

- Open QA queue → flagged labels with annotator name + reason
- Click label → split view: image left, details right
- Accept or reject → rejection opens comment field (required)
- Write rejection note → annotator notified, corrects & resubmits
- Coordinator approves → stats updated

Brainstorming Notes

- Split-screen QA — image and reviewer controls side-by-side, no navigation overhead
- Persistent shortcut overlay — collapsible panel pinned to the annotation screen
- Active class chip — high-contrast colour pill in the top bar to eliminate wrong-class errors
- Inline class guide — expandable description + example thumbnail without leaving the annotation screen
- Batch progress rings — circular indicators give more intuitive sense of completion than numbers
- Export validation report — green/red summary before download, no silent failures

4 · Design

Screen Inventory

#	Screen	Who	Status / Notes
1	Login / Registration	All	Built — split panel, tab switcher, role selector
2	Dashboard	All (role-adaptive)	Built — stat cards, project table, QA preview
3	Annotation Workspace	Annotator	Built — 4 states: Empty → Drawing → Selected → Saved
4	QA Review Interface	Coordinator	Built — 2 states: pending / decided
5	New Project Wizard	Researcher	Built — 3-step modal with live draft summary
6	Label Schema Editor	Researcher	Built — colour swatches, attribute definition cards
7	Dataset Import	Researcher	Built — drag-drop zone, tile status icons, batch assignment
8	Project Overview	Researcher / Coordinator	Built — 5-stat cards, annotator table, donut chart, timeline
9	Reports & Analytics	Researcher / Coordinator	Built — bar chart, IAA table, speed bars, rejection donut
10	Export Settings	Researcher	Built — 3 format cards, validation panel, live toggle
11	Profile & Settings	All	Built — profile form, notification toggles, shortcuts table

Key Design Decisions — Core Screens

Screen 1 — Login / Registration

- **Contextual Split:** Separates "Why" (marketing) from "How" (the form).
- **Unified Navigation:** Uses a tab switcher to reduce clicks and stay on one page.
- **Smart Forms:** Only shows extra fields when needed to keep sign-in minimal.
- **Instant Personalization:** Surfaces role selection early to customize the onboarding flow.

Screen 2 — Dashboard

- **Cleaner Navigation:** We split the menus between a sidebar and a top bar so the screen doesn't feel crowded.
- **Visual Categories:** We used distinct colors on stat cards to help users identify categories at a glance.
- **Balanced Layout:** A 64/36 split puts the focus on main projects while keeping the QA preview handy.
- **One-Click Decisions:** Users can resolve QA items right on the dashboard without having to open a new page.

Screen 3 — Annotation Workspace

- **Familiar UX:** Follows standard annotation layouts (like CVAT/Label Studio) for instant usability.
- **State-Based UI:** Uses 4 explicit visual states to guide the user through the annotation process.
- **Error Prevention:** Locks the attribute panel until an object is selected to keep data clean.
- **Contextual Shortcuts:** Displays hotkeys within tooltips to speed up the professional workflow.
- **Persistent Progress:** Keep users oriented with "Image X of Y" counters visible in two locations.

Screen 4 — QA Review Interface

- **Minimized Cognitive Load:** A centered layout helps users focus on one item rather than a complex dashboard.
- **Error Prevention:** "Pending" and "Decided" states lock the UI to ensure data accuracy.
- **Visual Triage:** Severity colors (Low confidence vs. Critical errors) let users assess items before reading the text.
- **Non-Linear Navigation:** A thumbnail strip at the bottom allows for quick jumping between any item in the queue.

Screens 5–7 — Project Wizard, Schema Editor, Dataset Import

- **Focused Sub-tasks:** Blurred backgrounds keep the user on-task without losing context.
- **Transparent Workflow:** A 3-step stepper manages user expectations from the start.
- **Real-time Feedback:** A live summary bar previews work before it's finalized.
- **Consistent Visuals:** Color swatches in the editor match the final annotation colors.
- **Detailed Upload Tracking:** Status icons show exactly which files are done and which are still processing.

Screens 8–10 — Project Overview, Reports, Export

- **5-stat card row:** Total Images, Annotated, QA Approved, In Review, Remaining — colour-coded for fast health check
- **QA rejection reasons as drill-through links:** Reports are an entry point into action, not a dead end
- **Format cards instead of a dropdown:** COCO JSON, YOLO TXT, Pascal VOC side-by-side — researcher can compare formats before selecting
- **Pre-export validation panel:** Issues surfaced and explained before a file is generated — no more silent export failures
- **Live export summary reacts to toggles:** Toggling 'Include low-confidence annotations' updates the annotation count in real time before committing to the download

5 · Test

Setup

We conducted 40-minute moderated 'think-aloud' sessions with three participants using a clickable Figma prototype.

Scenarios

Scenario A — Annotation Session:

"You're an annotator. Label all pedestrians in the first image with bounding boxes, tag age group and occlusion level, then move to the next image."

Scenario B — QA Review:

"You're the lab coordinator. Three annotations are flagged as low confidence. Review and reject the ones that are wrong, and leave a comment explaining why."

Scenario C — Export:

"You're the researcher. Your team finished 200 images. Export in COCO JSON and verify the export summary."

Observations

Scenario	What Worked	Problem	Severity	Fix
A	Box drawing intuitive in <30s. Shortcut N used unprompted after reading the overlay.	Attribute panel invisible until box selected → participant skipped attributes entirely.	High	Tooltip on first annotation: 'Select a box to add attributes'. Keep panel minimally visible always.
A	Progress bar updated immediately — motivated the annotator.	Active class chip not noticed → 2 images labeled with wrong class.	High	Larger chip + pulsing animation on first use. Warning on submit if class looks inconsistent.
B	Split-view praised by all: 'I can see everything without switching.'	Comment field not visibly required → participant clicked Reject without a comment and got confused.	Medium	Red border on comment field when Reject clicked before writing. Inline label: 'Required when rejecting'.
C	Export format selector clear. All participants picked COCO JSON correctly.	Warnings in validation report unexplained → participant unsure whether to proceed.	Medium	Expandable detail per warning with plain-language explanation and 'How to fix' link.
C	—	'Download All' button not found — participants hunted before finding per-format buttons.	Low	Prominent Download CTA below format selector that activates once format is chosen.

Summary

The main process works well, and everyone finished their tasks without help. However, we found two major issues: some important features were "hidden in plain sight." Our next step isn't to add more to the screen, but to make sure the right tools pop up at exactly the right time

Limitations & Next Steps

- Small participant count (n=3) — findings are directional, not statistically generalisable
 - No mobile support in v1 — mobile annotation is a common field use case
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